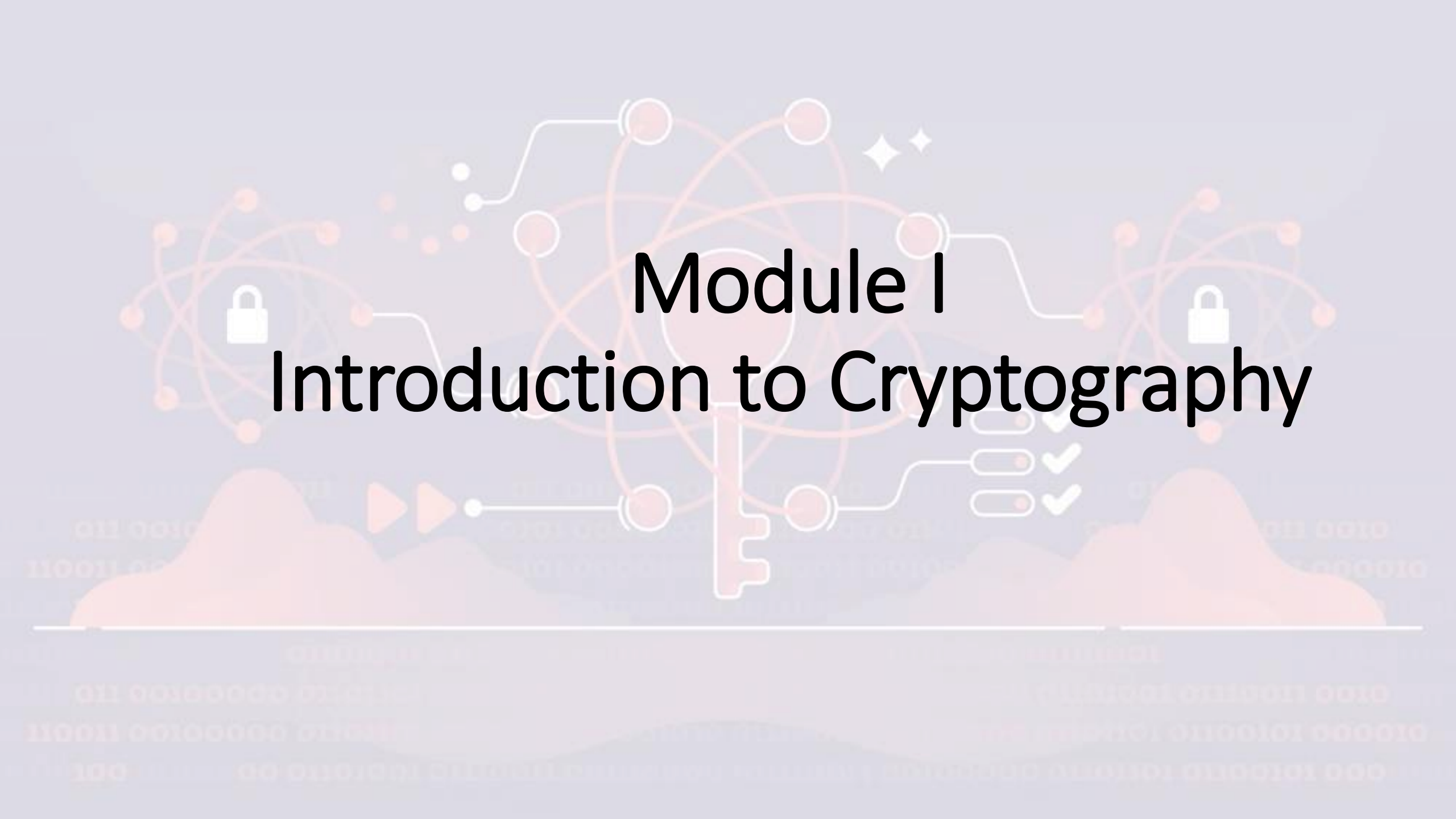


The background is a light purple gradient. It features several abstract elements: a central network of orange nodes connected by white lines, two white padlock icons on the left and right, a large white key icon in the center, and two green checkmarks on the right. At the bottom, there are stylized orange and pink clouds and a white horizontal line.

Module I

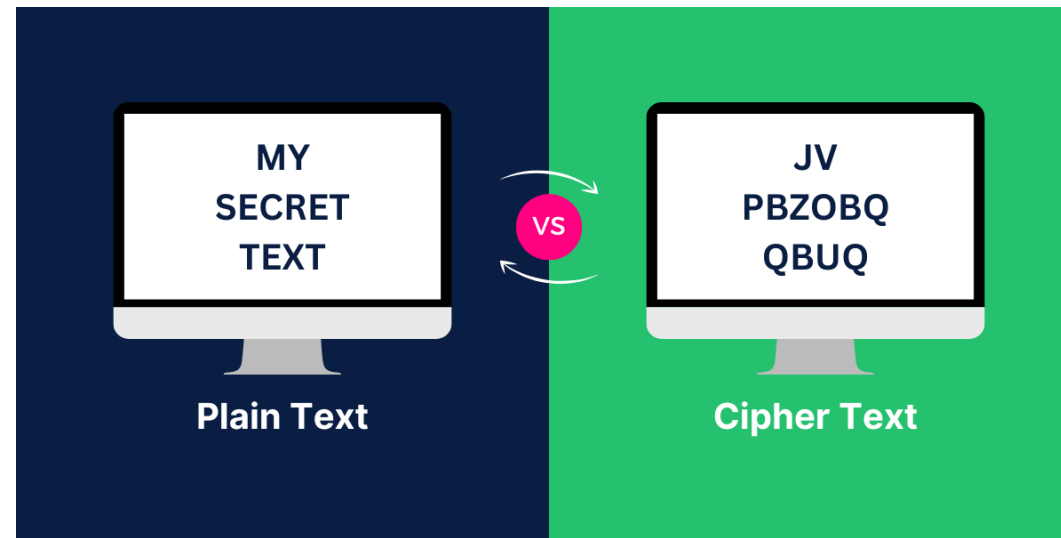
Introduction to Cryptography

What is Cryptography?

- Cryptography is a technique of **securing information and communications through the use of codes** so that only those persons for whom the information is intended can understand and process it. Thus, **preventing unauthorized access** to information.
- The prefix “**crypt**” means “**hidden**” and the suffix “**graphy**” means “**writing**”.
- In Cryptography, the techniques that are used to protect information are obtained from **mathematical concepts** and a **set of rule-based calculations** known as **algorithms** to convert messages in ways that make it hard to decode them.
- These algorithms are used for **cryptographic key generation, digital signing, and verification to protect data privacy, web browsing on the internet and to protect confidential transactions such as credit card and debit card transactions.**

Terminologies:

- **Plain Text:** The **original message** that the person wants to connect with the other is represented as Plain Text.
- **Cipher Text:** The message that cannot be learned by anyone or **meaningless message** is what it can call as Cipher Text. In Cryptography the original message is changed into non-readable message before the communication of actual message.

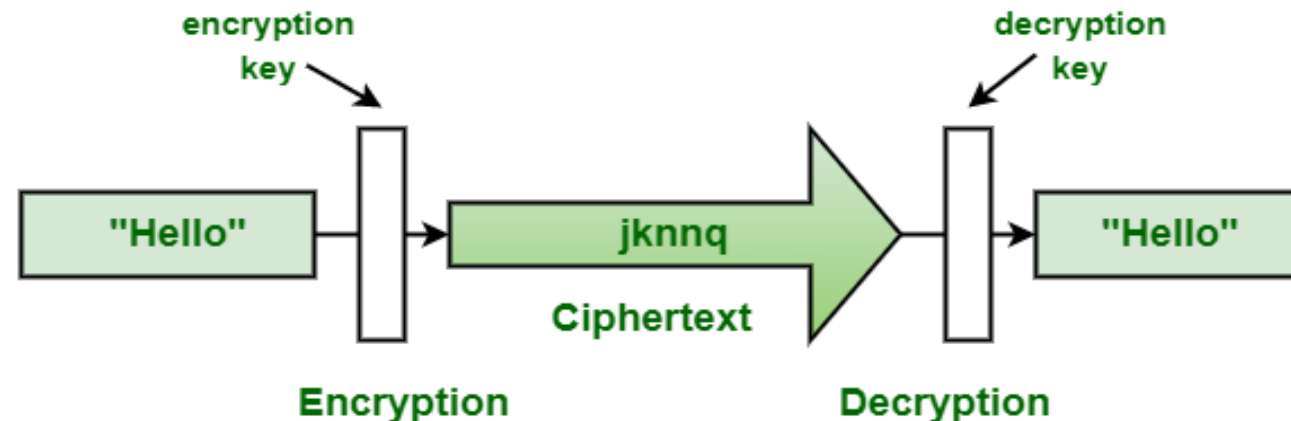


- **Encryption:**

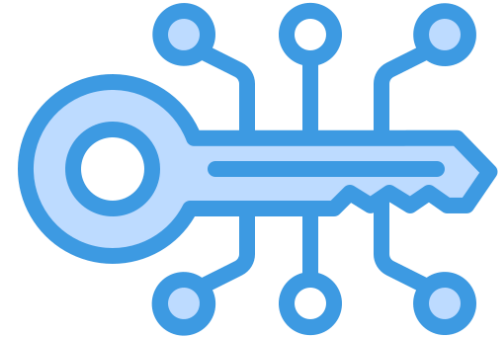
- A process of transforming **Plain Text into Cipher Text** is known as Encryption.
- Cryptography need the encryption approach to send confidential messages through an insecure channel.
- The phase of encryption requires two things such as **encryption algorithm and a key.**

- **Decryption:**

- A **reverse process** of encryption is known as Decryption.
- It is a procedure of transforming **Cipher Text into Plain Text.**
- Cryptography needs the decryption approach at the receiver side to acquire the original message from non-readable message (Cipher Text).
- The phase of decryption requires two things such as **Decryption algorithm and a key.**



- **Key:**
 - A Key is a **numeric or alpha numeric text or can be a unique symbol.**
 - The Key can be used at the time of encryption takes place on the Plain Text and at the time of decryption create place on the Cipher Text.
 - The selection of key in Cryptography is essential because the security of encryption algorithm depends precisely on it.



Features Of Cryptography

- **Confidentiality:** Information can only be accessed by the person for whom it is intended and no other person except him can access it.
- **Integrity:** Information cannot be modified in storage or transition between sender and intended receiver without any addition to information being detected.
- **Non-repudiation:** The creator/sender of information cannot deny his intention to send information at a later stage.
- **Authentication:** The identities of the sender and receiver are confirmed. As well destination/origin of the information is confirmed.
- **Interoperability:** Cryptography allows for secure communication between different systems and platforms.
- **Adaptability:** Cryptography continuously evolves to stay ahead of security threats and technological advancements.

Types Of Cryptography

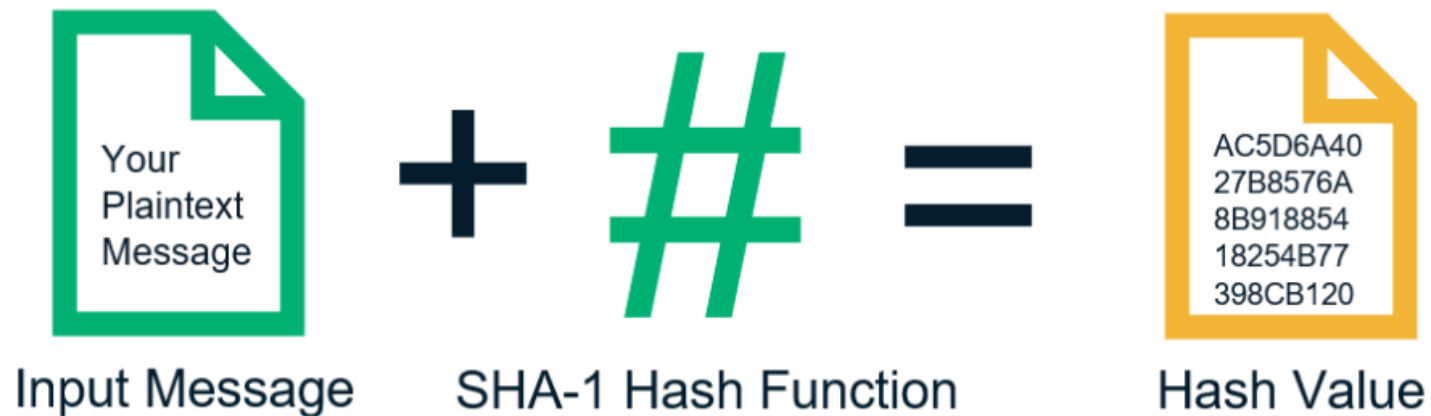
1. Symmetric Key Cryptography

- It is an encryption system where the **sender and receiver of a message use a single common key to encrypt and decrypt messages.**
- Symmetric Key Cryptography is **faster and simpler** but the problem is that the sender and receiver have to somehow exchange keys securely.
- The most popular symmetric key cryptography systems are:
 - **Data Encryption Systems (DES)**
 - **Advanced Encryption Systems(AES).**



2. Hash Functions

- There is no usage of any key in this algorithm.
- A hash value with a fixed length is calculated as per the plain text which makes it impossible for the contents of plain text to be recovered.
- operating systems use hash functions to encrypt passwords.



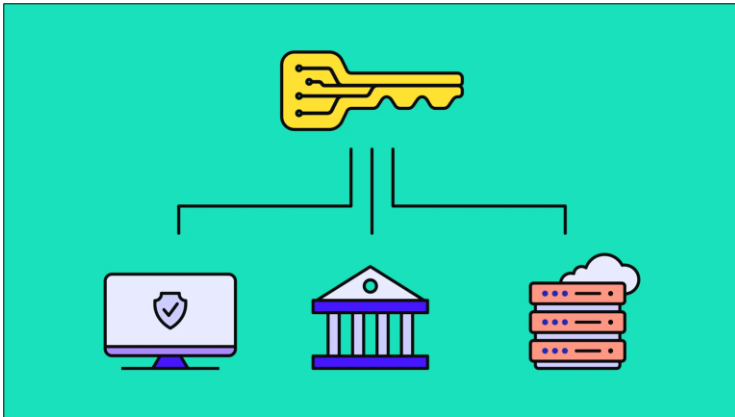
3. Asymmetric Key Cryptography

- In Asymmetric Key Cryptography a pair of keys is used to encrypt and decrypt information.
- A sender's public key is used for encryption and a receiver's private key is used for decryption.
- Public keys and Private keys are different.
- Even if the public key is known by everyone the intended receiver can only decode it because he alone knows his private key.
- The most popular asymmetric key cryptography algorithm is the RSA algorithm.



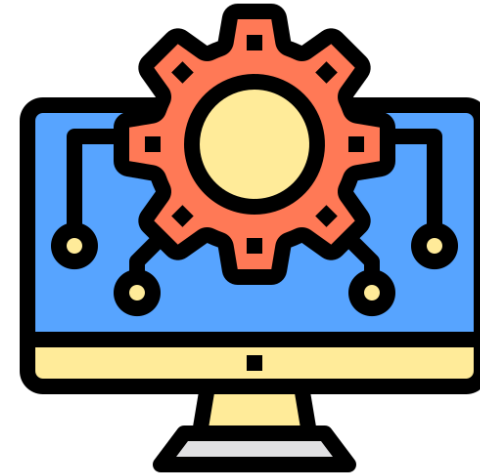
Applications of Cryptography

- Computer passwords
- Digital Currencies
- Secure web browsing
- Electronic Signatures
- Authentication
- Cryptocurrencies
- End-to-end Internet Encryption



Types of Cryptography Algorithms

- Advanced Encryption Standard (AES)
- Data Encryption Standard (DES)
- RSA
- Secure Hash Algorithm (SHA)



Advantages of Cryptography

- **Access Control:** Cryptography can be used for access control to ensure that ***only parties with the proper permissions have access to a resource***. Only those with the ***correct decryption key*** can access the resource thanks to encryption.
- **Secure Communication:** For secure online communication, cryptography is crucial. It offers secure mechanisms for transmitting private information like passwords, bank account numbers, and other sensitive data over the Internet.
- **Protection against attacks:** Cryptography aids in the defense against various types of assaults, including replay and man in the middle attacks. It offers strategies for spotting and stopping these assaults.
- **Compliance with legal requirements:** Cryptography can assist firms in meeting a variety of legal requirements, including data protection and privacy legislation.

The OSI Security Architecture

- The OSI Security Architecture is internationally recognized and provides a standardized technique for ***deploying security measures within an organization***.
- ITU-T Recommendation X.800, Security Architecture for OSI, defines such a systematic approach.
- The OSI security architecture is useful to managers as a way of organizing the task of providing security.

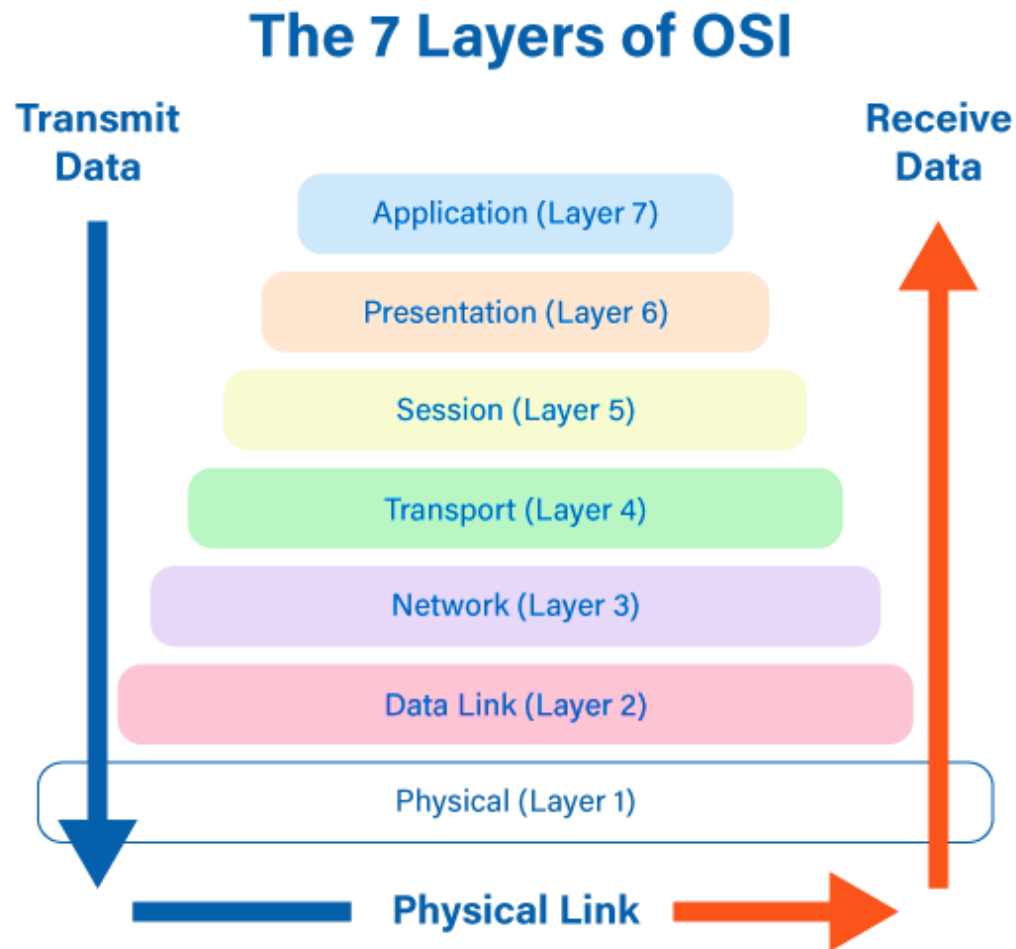
○ **OSI : Open Systems Interconnection**

○ **ITU : The International Telecommunication Union (ITU)**

○ **Telecommunication Standardization Sector (ITU-T) is a United Nations sponsored agency that develops standards.**

What is the OSI Model?

- The OSI model can be considered a *universal language for computer networking*.
- It is based on the concept of *divide and conquer*, it *splits up the communication system into 7 abstract layers, and the layer is stacked upon the previous layer*.



Classification of OSI Security Architecture

- The OSI security architecture focuses on *security attacks, mechanisms, and services*.
- These can be defined briefly as follows:
 - **Security attack:** Any action that compromises the security of information owned by an organization.
 - **Security mechanism:** A process (or a device incorporating such a process) that is *designed to detect, prevent, or recover from a security attack*.
 - **Security service:** A processing or communication service that enhances the *security of the data processing systems and the information transfers of an organization*. The services are intended to counter security attacks, and they make use of one or more security mechanisms to provide the service.

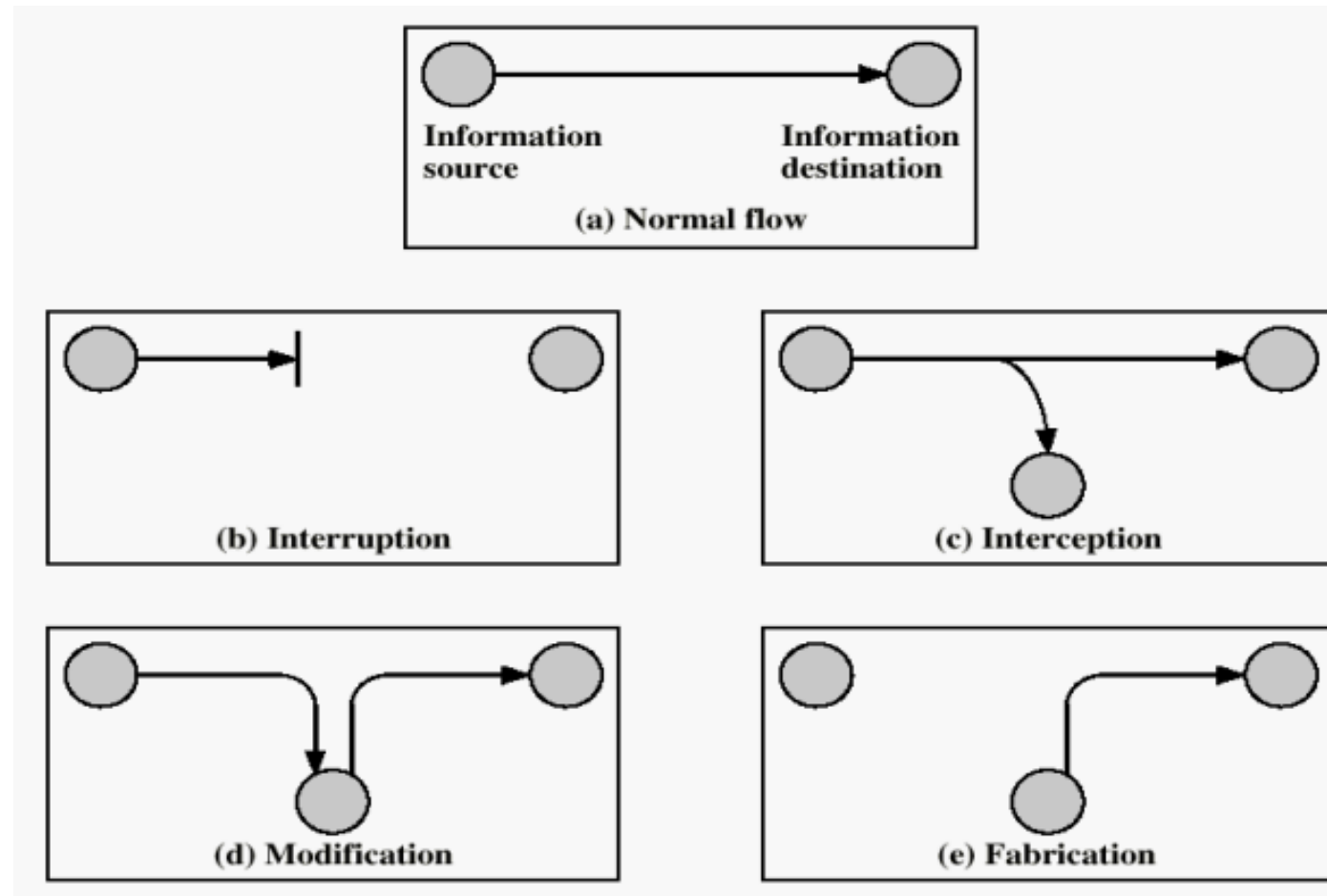
Security violations:

1. When data is transmitted from one system to other, the data would be access by another user.
2. An unauthorized user may access and change the original message (data).
3. Duplicate message may be sent by unauthorized user by discarding the original message.
4. Delaying the message during transmission.
5. Denying about sent message, i.e. user who sent the message itself can deny it.



Security Threats

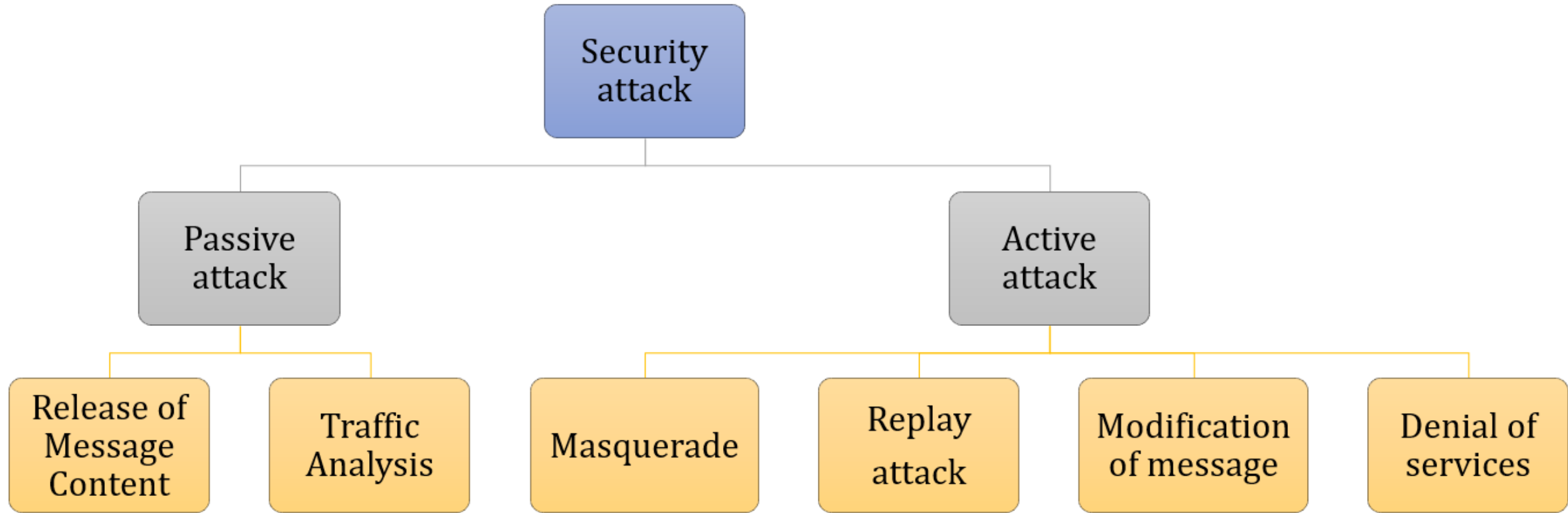
- A potential for ***violation of security***, which exists when there is a circumstance, capability, action, or event that could ***breach security and cause harm***.
- That is, ***a threat is a possible danger that might exploit a vulnerability***.
- Security Threats:



- **Interruption:** an asset of the system is *destroyed or becomes unavailable*. This is an *attack on unavailability*. E.g: cutting the communication line.
- **Interception:** an unauthorized user *gain access to an asset*. This is an attack on *confidentiality*. Eg: wiretapping.
- **Modification:** an unauthorized user *not only gains access to, but tampers with an asset*. This is an *attack on integrity*. Eg: changing values in data file.
- **Fabrication:** an unauthorized user *inserts counterfeit objects into the system*. This is an attack on *authenticity*. Eg: insertion of spurious messages.

I. Security Attacks

- In computer networks, an attack is any ***attempt to destroy, expose, alter, disable, steal or gain unauthorized access to or make unauthorized use of an asset.***
- Types of attacks: ***passive attacks and active attacks.***
 - A passive attack attempts to ***learn or make use of information from the system but does not affect system resources.***
 - An active attack ***attempts to alter system resources or affect their operation.***

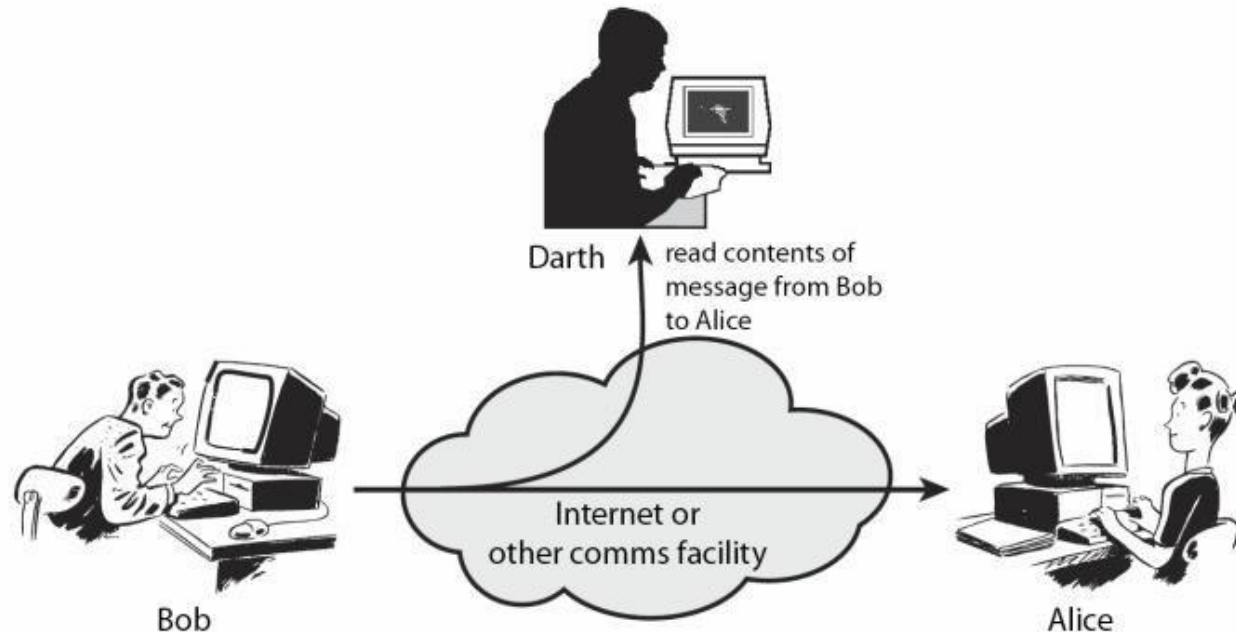


1. Passive attack:

- This attack is just **accessing the message and not altering the message.**
- Passive attacks are in the nature of **eavesdropping on, or monitoring of, transmissions.**
- The **goal of the opponent is to obtain information that is being transmitted.**
- Two types of passive attacks are
 - **Release of message contents**
 - **Traffic analysis.**

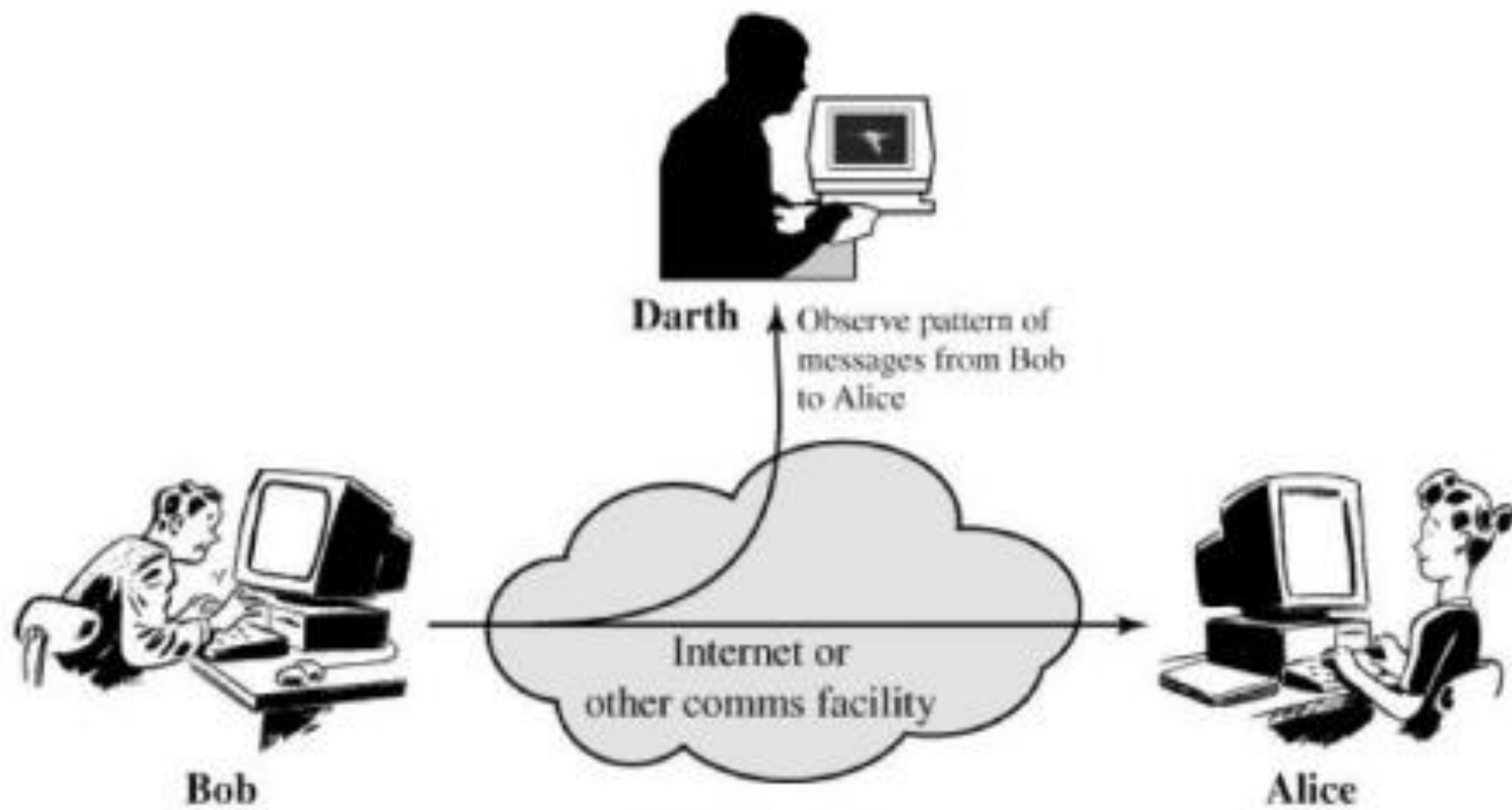
✓ Release of message contents:

- The release of message contents is easily understood.
- **To read the contents of message.**
- A telephone conversation, an electronic mail message, and a transferred file may contain sensitive or confidential information.
- Techniques: Packet Sniffing , man-in-the-middle attacks.



✓ Traffic analysis:

- Monitoring of unprotected communications, decrypting weakly encrypted traffic, and ***capturing authentication information such as passwords.***
- The common technique for ***masking contents is encryption.***
- If we had encryption protection in place, ***an opponent might still be able to observe the pattern of these messages.***
- The opponent ***could determine the location and identity of communicating hosts and could observe the frequency and length of messages being exchanged.***
- This information might be useful in ***guessing the nature of the communication*** that was taking place.
- Passive attacks are ***very difficult to detect*** because they ***do not involve any alteration*** of the data.
- Techniques: network flow analysis, or protocol analysis.

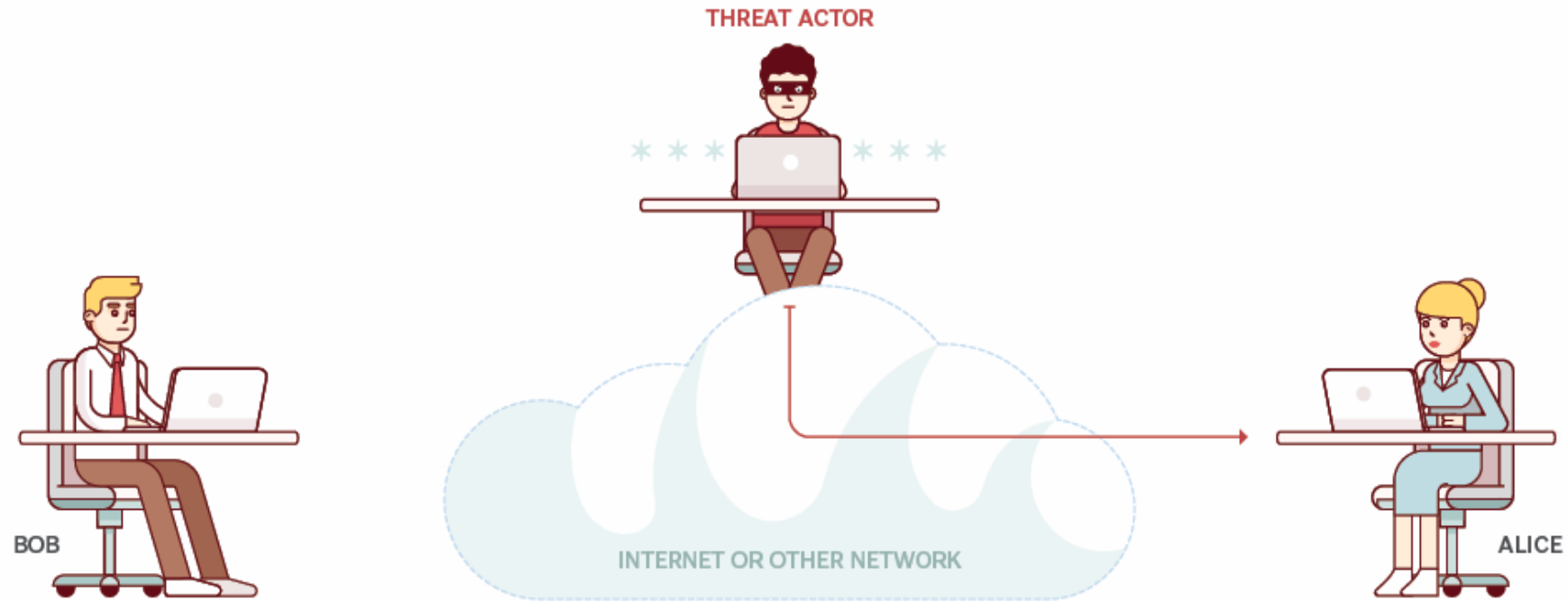


2. Active attacks:

- Active attacks *involve some modification of the data stream or the creation of a false stream* and can be subdivided into four categories:
 - ✓ Masquerade
 - ✓ Replay
 - ✓ Modification of messages
 - ✓ Denial of service

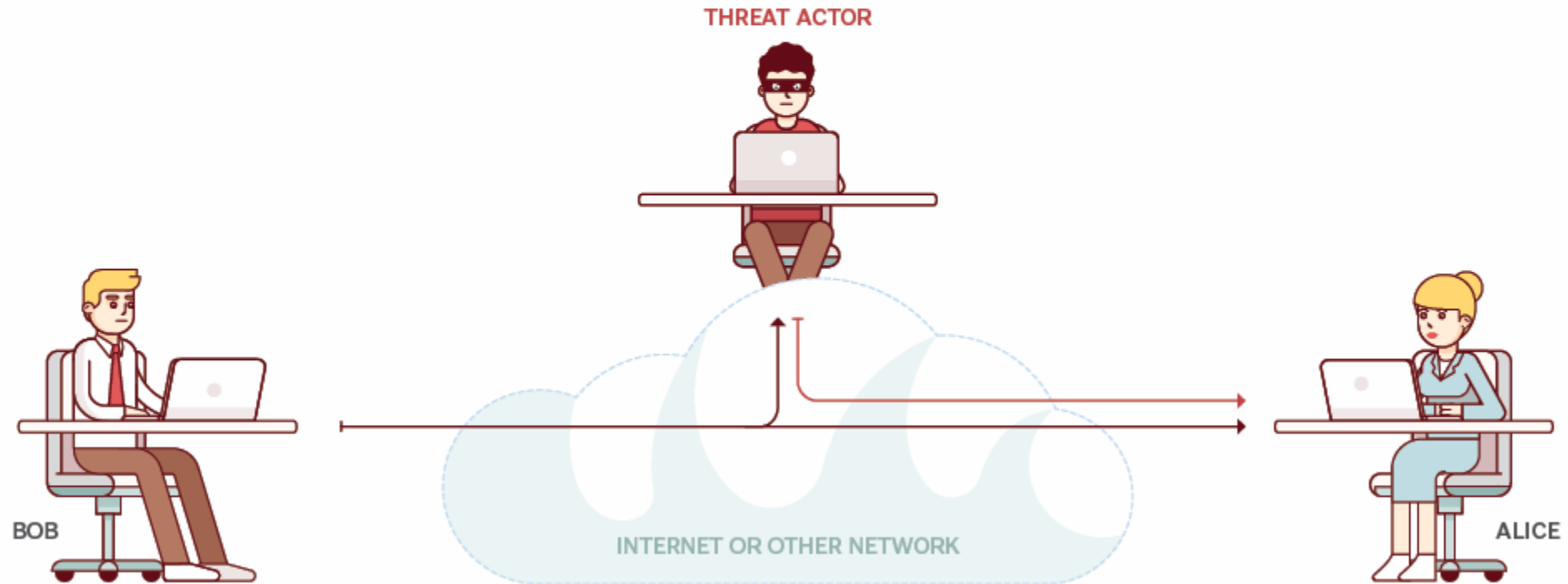
Masquerade attack

In an email masquerade attack, the attacker uses stolen login information or access via another security gap to send a message to Alice that looks like it's from Bob.



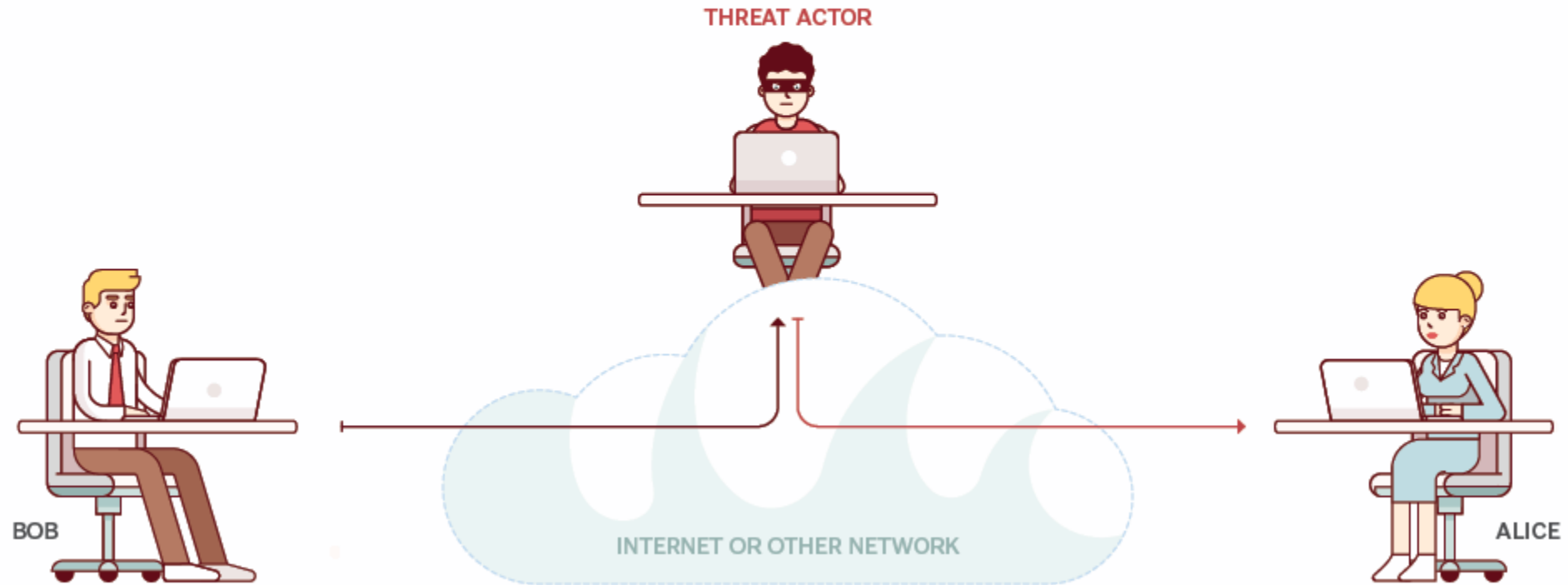
Hijacking or replay attack

In an email replay or hijacking attack, the attacker uses nefarious means to capture a message Bob sent to Alice and then replays it to get information or access from Alice, who thinks she's communicating with Bob.



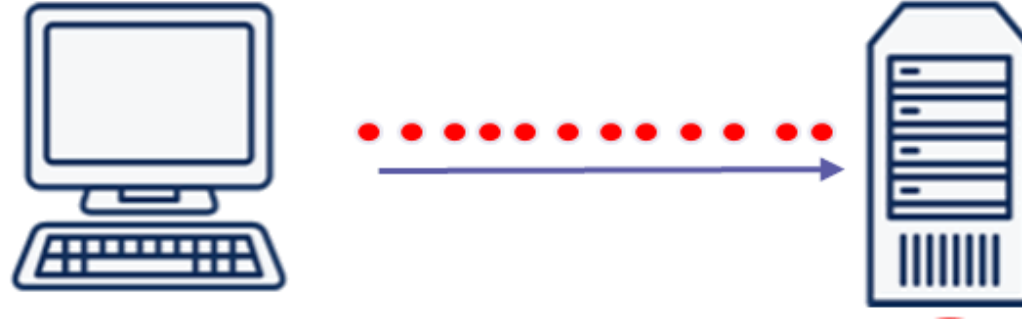
Message modification attack

In a message modification attack, the attacker captures a message from Bob, changes it and then sends it on to Alice, who thinks the message is from Bob.



DoS attack

- Denial of Service attacks involve the ***attacker sending a large volume of traffic to a system, network, or device in an attempt to overwhelm it and make it unavailable to users.***



- ***Two Techniques:***

- **Flooding:** The attacker ***floods the target computer with internet traffic*** to the point that the traffic overwhelms the target system. The ***target system is unable to respond*** to any requests or process any data, ***making it unavailable*** to legitimate users.
- **Malformed data:** ***Rather than overloading*** a system with requests, an attacker may strategically ***send data that a victim's system cannot handle***. For example, a DoS attack could corrupt system memory, manipulate fields in the network protocol packets or exploit servers.

Signs of a DoS attack



Degradation in network performance

Especially when attempting to open files stored on the network or when accessing websites.



Specific website unavailable

A particular site does not open or cannot be found.



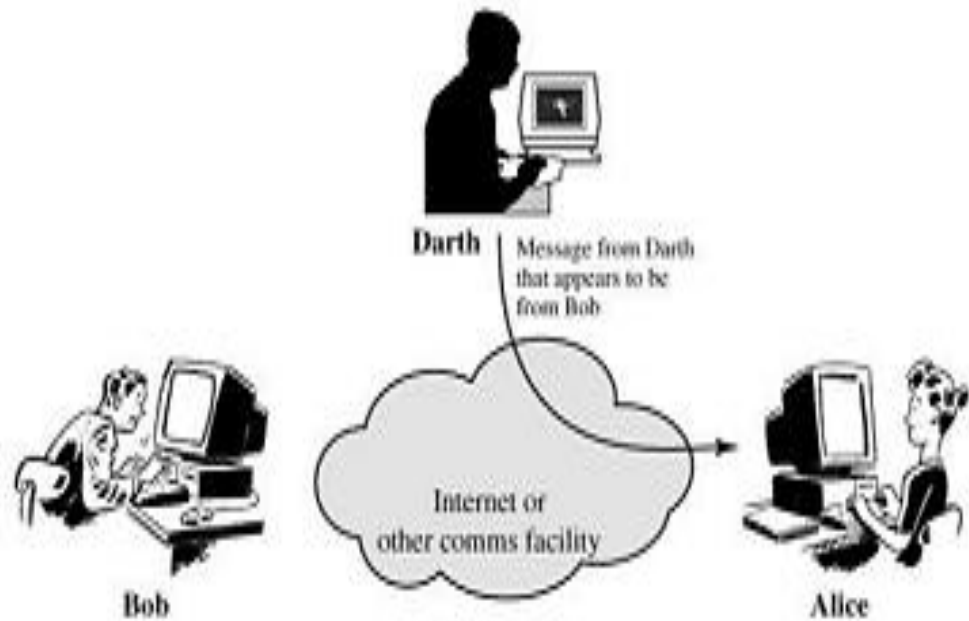
Inability to access any website

All websites are inaccessible on the network.

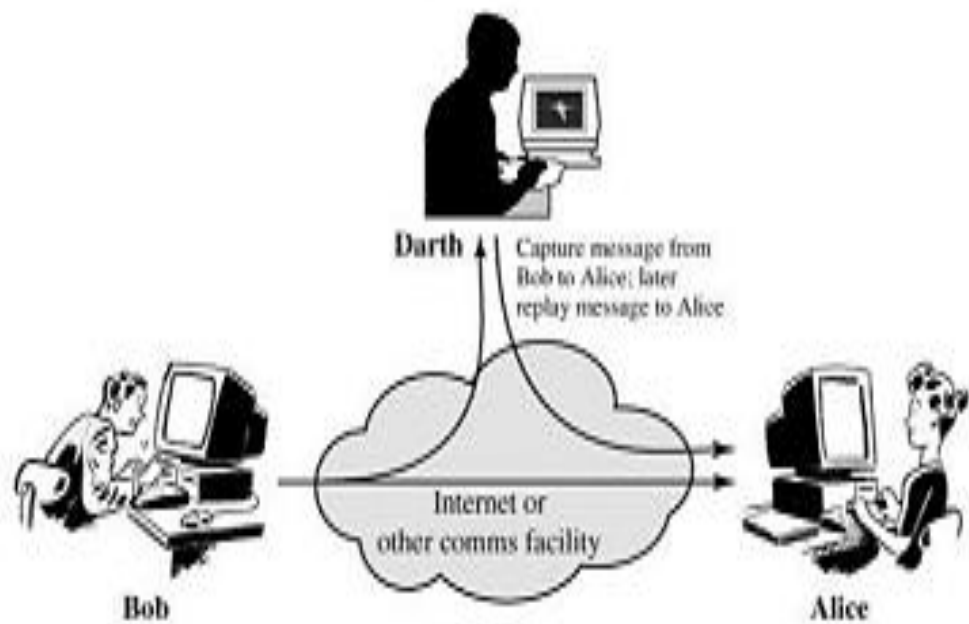


High volume of email spam

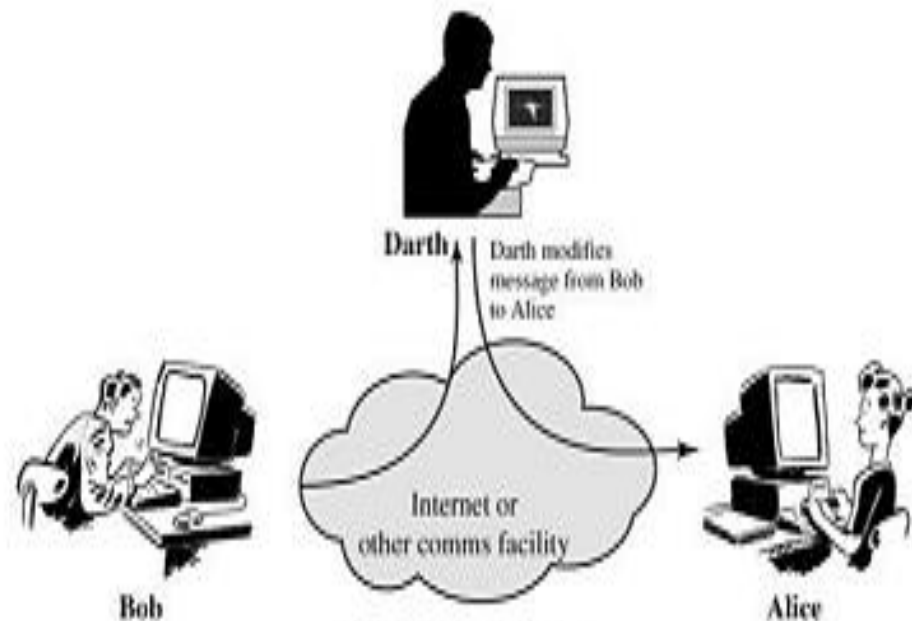
A higher than usual volume of spam email.



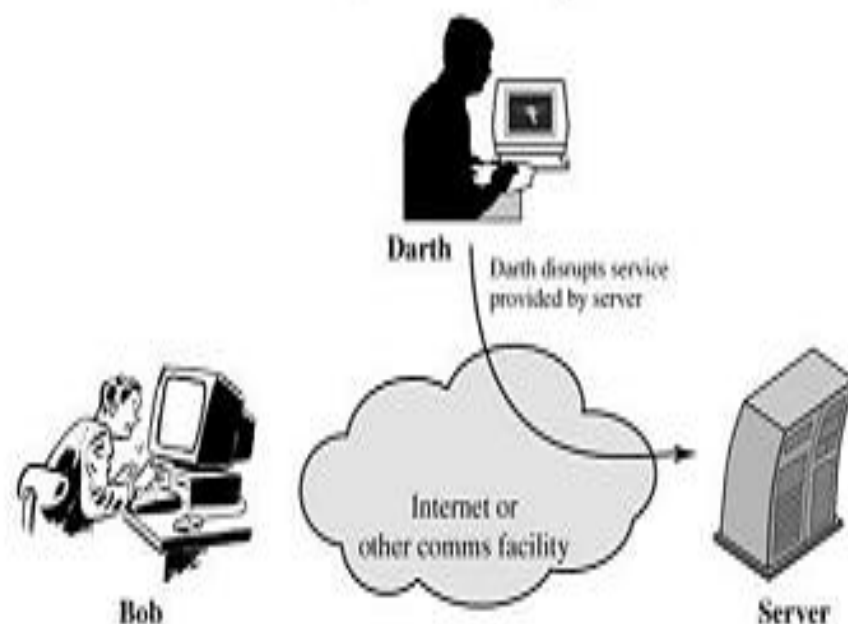
(a) Masquerade



(b) Replay



(c) Modification of messages



(d) Denial of service

II. Security Mechanism

- The mechanism that is ***built to identify any breach of security or attack on the organization, is called a security mechanism.***
- Security Mechanisms are also ***responsible for protecting a system, network, or device against unauthorized access, tampering, or other security threats.***
- The security mechanisms are divided into those that are implemented in a ***specific protocol layer*** and those that are ***not specific*** to any particular protocol layer or security service.
- Security mechanisms defined in X.800
 1. **Specific Security Mechanisms**
 2. **Pervasive Security Mechanisms**

- **Specific Security Mechanisms:** May be incorporated into the appropriate protocol layer in order to provide some of the OSI security services.
 1. **Encipherment (Encryption):** The use of mathematical algorithms to transform data into a form that is not readily intelligible.
 2. **Digital Signature:** A digital signature is a mathematical technique used to validate the authenticity and integrity of a message, software, or digital document.
 3. **Access Control:** A variety of mechanisms that grants or denies access to resources based on user identity, role or privileges.
 4. **Data Integrity:** A variety of mechanisms used to assure the integrity of a data unit or stream of data units. That prevent unauthorized modifications, deletions or alterations of data.
 5. **Authentication Exchange:** A mechanism intended to ensure the identity of an entity by means of information exchange.
 6. **Traffic Padding:** The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts.
 7. **Routing Control:** Refers to the process of controlling and managing the flow of data packets between networks, ensuring that data is transmitted securely and efficiently.
 8. **Notarization:** The use of a trusted third party to assure certain properties of a data exchange.

- **Pervasive Security Mechanisms:** Mechanisms that is not specific to any particular OSI security service or protocol layer.
- 1. **Trusted Functionality:** Ability of a system or device to perform specific security related functions in a trustworthy and reliable manner. (e.g., as established by a security policy).
- 2. **Security Label:** The marking bound to a resource (which may be a data unit) that names or designates the security attributes of that resource.
- 3. **Event Detection:** Detection of security-relevant events.
- 4. **Security Audit Trail:** Data collected and potentially used to facilitate a security audit, which is an independent review and examination of system records and activities.
- 5. **Security Recovery:** Deals with requests from mechanisms, such as event handling and management functions, and takes recovery actions.

III. Security services

- X.800 defines a security service as a service provided by a protocol layer of communicating open systems, which ***ensures adequate security of the systems or of data transfers.***
- X.800 divides these services into ***five categories and fourteen specific services***
 - 1. Authentication (who created or sent the data)***
 - 2. Access control / Availability (prevent misuse of resources)***
 - 3. Confidentiality (privacy)***
 - 4. Integrity (has not been altered)***
 - 5. Non-repudiation (the order is final)***

1. Authentication - assurance that the communicating entity is the one that it claims to be. The process of proving one's identity.

- **Peer Entity Authentication:** Used in association with a logical connection to provide confidence in the identity of the entities connected.
- **Data Origin Authentication:** In a connectionless transfer, provides assurance that the source of received data is as claimed.

2. Access Control/Availability - prevention of the unauthorized use of a resource

3. Data Confidentiality –protection of data from unauthorized disclosure.

- **Privacy** - Ensuring that no one can read the message except the intended receiver.
- **Connection Confidentiality**-The protection of all user data on a connection.
- **Connectionless Confidentiality**-The protection of all user data in a single data block
- **Selective-Field Confidentiality**-The confidentiality of selected fields within the user data on a connection or in a single data block.
- **Traffic Flow Confidentiality** -The protection of the information that might be derived from observation of traffic flows.

4. Data Integrity - The assurance that data received are exactly as sent by an authorized entity (i.e., contain no modification, insertion, deletion, or replay). Assuring the receiver that the received message has not been altered in any way from the original.

- **Connection Integrity with Recovery:** Provides for the integrity of all user data on a connection and detects any modification, insertion, deletion, or replay of any data within an entire data sequence, with recovery attempted.
- **Connection Integrity without Recovery:** As above, but provides only detection without recovery.
- **Selective-Field Connection Integrity:** Provides for the integrity of selected fields within the user data of a data block transferred over a connection and takes the form of determination of whether the selected fields have been modified, inserted, deleted, or replayed.
- **Connectionless Integrity:** Provides for the integrity of a single connectionless data block and may take the form of detection of data modification. Additionally, a limited form of replay detection may be provided.
- **Selective-Field Connectionless Integrity:** Provides for the integrity of selected fields within a single connectionless data block; takes the form of determination of whether the selected fields have been modified.

5. Non-Repudiation : Protection against denial by one of the parties in a communication.

A mechanism to prove that the sender really sent this message.

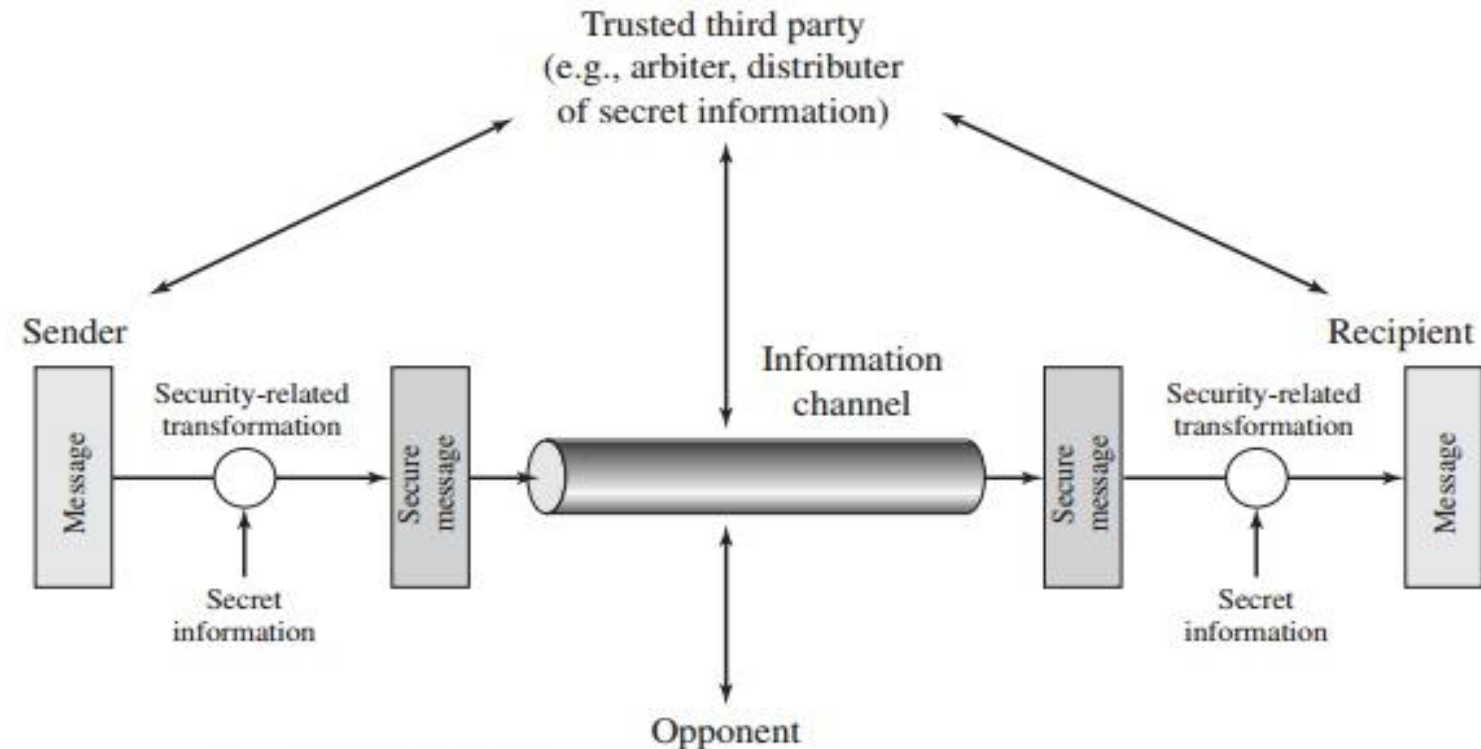
- **Non-repudiation, Origin** -Proof that the message was sent by the specified party.
- **Non-repudiation, Destination** -Proof that the message was received by the specified party.

Benefits of OSI Security Architecture

- **Providing Security:** OSI Architecture in an organization provides the needed security and safety, preventing potential threats and risks.
- **Organizing Task:** The OSI architecture makes it easy for managers to build a security model for the organization based on strong security principles.
- **Meets International Standards:** Security services are defined and recognized internationally meeting international standards.
- **Interoperability:** By dividing network functions into multiple levels, the OSI model makes it easier for different hardware and software components to work together.
- **Scalability:** The layered method makes networks scalable. New technologies and protocols can be seamlessly added without interrupting the overall system.
- **Flexibility:** Each layer can evolve separately, providing flexibility for technology and application changes.

A Model for Network Security

- A message is to be transferred from one party to another across some sort of internet.
- The two parties, who are the **principals in this transaction**, must cooperate for the exchange to take place.
- A **logical information channel** is established by defining a route through the internet from source to destination and by the cooperative use of **communication protocols** (e.g., TCP/IP) by the two principals.



- Security aspects come into play when it is necessary or desirable to **protect the information transmission from an opponent who may present a threat** to confidentiality, authenticity, and so on.

All the techniques for providing security have two components:

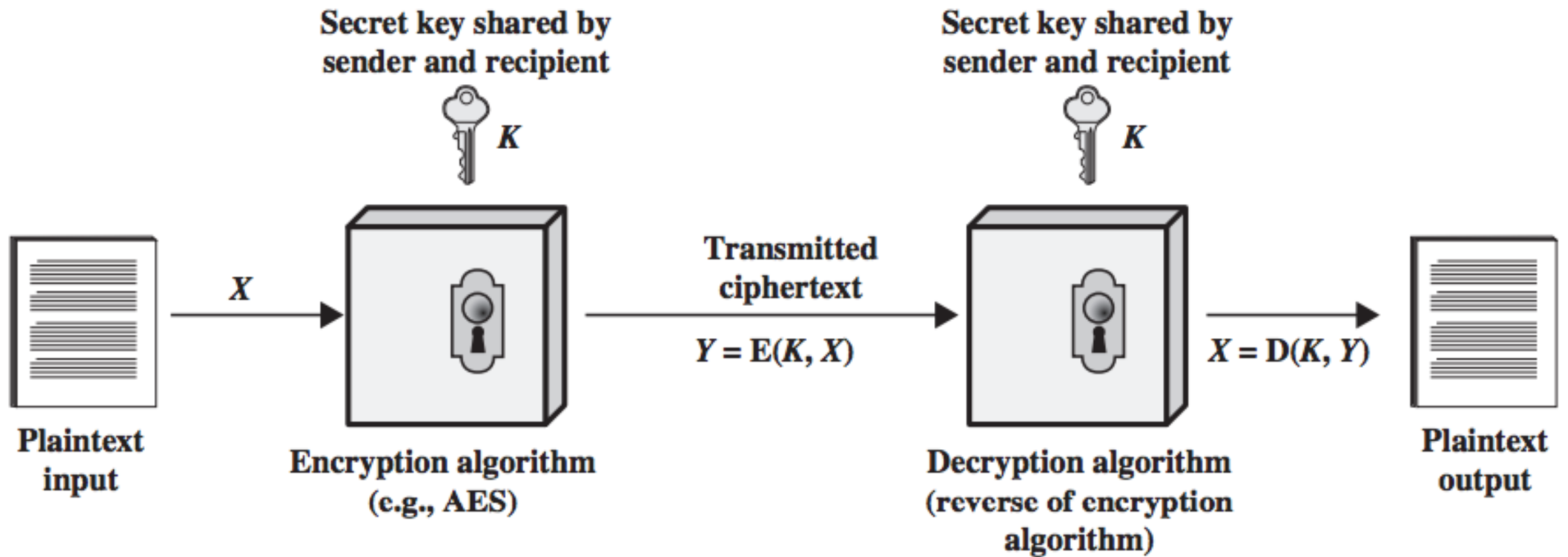
- **A security-related transformation on the information to be sent.** Examples include the encryption of the message, which scrambles the message so that it is unreadable by the opponent, and the addition of a code based on the contents of the message, which can be used to verify the identity of the sender.
- **Some secret information shared by the two principals and, it is hoped, unknown to the opponent.** An example is an encryption key used in conjunction with the transformation to scramble the message before transmission and unscramble it on reception.
- **A trusted third party** may be needed to achieve secure transmission. For example, a third party may be **responsible for distributing the secret information** to the two principals while keeping it from any opponent. Or a third party may be **needed to arbitrate disputes between the two principals** concerning the authenticity of a message transmission.

- This general model shows that there are **four basic tasks in designing a particular security service**:
 - ✓ **Design an algorithm** for performing the security-related transformation. The algorithm should be such that an opponent cannot defeat its purpose.
 - ✓ **Generate the secret information** to be used with the algorithm.
 - ✓ **Develop methods** for the distribution and sharing of the secret information.
 - ✓ **Specify a protocol** to be used by the two principals that makes use of the security algorithm and the secret information to achieve a particular security service.

SYMMETRIC CIPHER MODEL

A symmetric encryption scheme has five ingredients

- **Plaintext:** This is the original intelligible message or data that is fed into the algorithm as input.
- **Encryption algorithm:** The encryption algorithm performs various substitutions and transformations on the plaintext.
- **Secret key:** The secret key is also input to the encryption algorithm. The key is a value independent of the plaintext and of the algorithm. The algorithm will produce a different output depending on the specific key being used at the time. The exact substitutions and transformations performed by the algorithm depend on the key.
- **Ciphertext:** This is the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts. The ciphertext is an apparently random stream of data and, as it stands, is unintelligible.
- **Decryption algorithm:** This is essentially the encryption algorithm run in reverse. It takes the ciphertext and the secret key and produces the original plaintext.



Simplified Model of Conventional Encryption

- Here the original message, referred to as plaintext, is converted into apparently random nonsense, referred to as cipher text.
- The encryption process consists of an algorithm and a key.
- The key is a value independent of the plaintext.
- Changing the key changes the output of the algorithm.
- Once the cipher text is produced, it may be transmitted.
- Upon reception, the cipher text can be transformed back to the original plaintext by using a decryption algorithm and the same key that was used for encryption.
- The security depends on several factors.
- First, the encryption algorithm must be powerful enough that it is impractical to decrypt a message on the basis of cipher text alone.
- Beyond that, the security depends on the secrecy of the key, not the secrecy of the algorithm.

Two requirements for secure use of symmetric encryption: –

- A strong encryption algorithm
- A secret key known only to sender / receiver
- $Y = EK(X)$
- $X = DK(Y)$
- assume encryption algorithm is known
- implies a secure channel to distribute key A source produces a message in plaintext, $X = [X1, X2... XM]$ where M are the number of letters in the message.
- A key of the form $K = [K1, K2... KJ]$ is generated.
- If the key is generated at the source, then it must be provided to the destination by means of some secure channel.

- With the message X and the encryption key K as input, the encryption algorithm forms the cipher text $Y = [Y_1, Y_2, Y_N]$.
- This can be expressed as $Y = EK(X)$ The intended receiver, in possession of the key, is able to invert the transformation: $X = DK(Y)$
- An opponent, observing Y but not having access to K or X , may attempt to recover X or K or both. It is assumed that the opponent knows the encryption and decryption algorithms.
- If the opponent is interested in only this particular message, then the focus of effort is to recover X by generating a plaintext estimate.
- Often if the opponent is interested in being able to read future messages as well, in which case an attempt is made to recover K by generating an estimate.

CLASSICAL ENCRYPTION TECHNIQUES

- Symmetric encryption is a form of cryptosystem in which encryption and decryption are performed using the same key. It is also known as conventional encryption.
- Symmetric encryption transforms plaintext into ciphertext using a secret key and an encryption algorithm. Using the same key and a decryption algorithm, the plaintext is recovered from the ciphertext.
- The two types of attack on an encryption algorithm are **cryptanalysis**, based on properties of the encryption algorithm, and **brute-force**, which involves trying all possible keys.
- Traditional (pre-computer) symmetric ciphers use substitution and/or transposition techniques. Substitution techniques map plaintext elements (characters, bits) into ciphertext elements. Transposition techniques systematically transpose the positions of plaintext elements.
- Rotor machines are sophisticated pre-computer hardware devices that use substitution techniques.
- Steganography is a technique for hiding a secret message within a larger one in such a way that others cannot discern the presence or contents of the hidden message.

CLASSICAL ENCRYPTION TECHNIQUES

There are two basic building blocks of all encryption techniques:

- **Substitution**
- **Transposition**

SUBSTITUTION TECHNIQUES

- A substitution technique is one in which the **letters of plaintext are replaced by other letters or by numbers or symbols.**
- If the plaintext is viewed as a **sequence of bits**, then substitution involves **replacing plaintext bit patterns with cipher text bit patterns.**

1. Caesar cipher (or) shift cipher

- The earliest known use of a substitution cipher, and the simplest, was by **Julius Caesar**.
- The Caesar cipher involves **replacing each letter of the alphabet with the letter standing three places further down the alphabet**.
- For example:

plain: meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

- Note that the alphabet is wrapped around, so that the letter following Z is A. We can define the transformation by listing all possibilities, as follows:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c

- Let us assign a numerical equivalent to each letter:

a	B	C	d	e	f	g	h	i	j	K	L	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	O	P	q	r	s	t	u	v	w	X	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Then the algorithm can be expressed as follows.

- For each plaintext letter P, substitute the ciphertext letter C

$$C = E(3, P) = (P + 3) \bmod 26$$

- A shift may be of any amount, so that the general Caesar algorithm is

$$C = E(k, P) = (p + k) \bmod 26$$

- where k takes on a value in the range 1 to 25.
- The decryption algorithm is simply

$$P = D(k, C) = (C - k) \bmod 26$$

- If it is known that a given ciphertext is a Caesar cipher, then a **brute-force** cryptanalysis is easily performed: Simply try all the 25 possible keys.
- Three important characteristics of this problem enabled us to use a brute-force cryptanalysis:
 - **The encryption and decryption algorithms are known.**
 - **There are only 25 keys to try.**
 - **The language of the plaintext is known and easily recognizable.**

2. Playfair cipher

- The best-known **multiple letter encryption cipher** is the Playfair, which treats **diagrams** in the plaintext as single units and translates these units into cipher text diagrams.
- The playfair algorithm is based on the use of **5x5 matrix of letters** constructed **using a keyword**.
- Let the keyword be “monarchy”.
- The matrix is constructed by filling in the letters of the keyword (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetical order.
- The letter “l” and “j” count as one letter.
- Plaintext is encrypted two letters at a time According to the following rules:
- Repeating plaintext letters that would fall in the same pair are separated with a **Filler letter** such as “x”.

- Plaintext letters that fall in the **same row** of the matrix are each **replaced by the letter to the right**, with the first element of the row following the last.
 - **Same Row- [→] wrap Around**
- Plaintext letters that fall in the **same column** are **replaced by the letter beneath**, with the top element of the column following the last.
 - **Same Column- [↓] wrap Around**
- Otherwise, each plaintext letter is replaced by the letter that lies in its own row **And the column occupied by the other plaintext letter.**

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

- Plaintext = **meet me at the school house**
- Splitting two letters as a unit => me et me at th es ch ox ol ho us ex
Corresponding cipher text => CL KL CL RS PD IL HY AV MP HF XL IU

Strength of playfair cipher

- Playfair cipher is a great advance over simple mono alphabetic ciphers.
- Since there are 26 letters, $26 \times 26 = 676$ diagrams are possible, so identification of individual diagram is more difficult.

3. Hill Cipher

- Another interesting multi letter cipher is the Hill cipher, developed by the mathematician Lester Hill in 1929.
- The encryption algorithm takes m successive plaintext letters and substitutes for them m ciphertext letters.
- The substitution is determined by m linear equations in which each character is assigned a numerical value ($a = 0, b = 1 \dots z = 25$).
- For $m = 3$, the system can be described as follows:

$$c1 = (k11P1 + k12P2 + k13P3) \bmod 26$$

$$c2 = (k21P1 + k22P2 + k23P3) \bmod 26$$

$$c3 = (k31P1 + k32P2 + k33P3) \bmod 26$$

This can be expressed in term of column vectors and matrices:

$$\begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} \text{mod } 26$$

$$C = KP \text{ mod } 26$$

$$P = K^{-1}C \text{ mod } 26 = KK^{-1}P = P$$

where C and P are column vectors of length 3, representing the plaintext and ciphertext, and K is a 3 x 3 matrix, representing the encryption key. Operations are performed mod 26.

4. Polyalphabetic Ciphers

- Another way to improve on the simple monoalphabetic technique is to use different monoalphabetic substitutions as one proceeds through the plaintext message.
- The general name for this approach is **polyalphabetic substitution cipher**.
- All these techniques have the following features in common:
 - A set of related monoalphabetic substitution rules is used.
 - A key determines which particular rule is chosen for a given transformation.
- To aid in understanding the scheme and to aid in its use, a **matrix** known as the **Vigenère tableau** is constructed.
- Each of the 26 ciphers is laid out horizontally, with the key letter for each cipher to its left.
- A normal alphabet for the plaintext runs across the top.

- The process of encryption is simple:

Given a key letter x and a plaintext letter y, the ciphertext letter is at the intersection of the row labeled x and the column labeled y; in this case the ciphertext is V.

		Plaintext																									
Key		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
	B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
	C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
	D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
	E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
	F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
	G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
	H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
	I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
	J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
	K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
	L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
	M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
	N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
	O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
	W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
	X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
	Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

- To encrypt a message, a key is needed that is as long as the message.
- Usually, the key is a repeating keyword.
- For example, if the keyword is deceptive, the message "**we are discovered save yourself**" is encrypted as follows:

key: deceptive

plaintext: wearediscoveredsaveyourself

ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

- Decryption is equally simple. The key letter again identifies the row.
- The position of the ciphertext letter in that row determines the column, and the plaintext letter is at the top of that column.

TRANSPOSITION TECHNIQUES

- A very different kind of mapping is achieved by performing some sort of **permutation on the plaintext** letters. This technique is referred to as a **transposition cipher**.

1. Rail Fence Transposition

- Rail-Fence is a basic Transposition method in which **plaintext is written as a series of diagonals, which is then read row by row to generate the ciphertext**.
- The simplest such cipher is the rail fence technique.
- For example, to encipher the message "**meet me after the toga party**" with a rail fence of depth 2, we write the following:

m e m a t r h t g p r y
e t e f e t e o a a t

- The encrypted message is **MEMATRHTGPRYETEFETEOAAT**
- This sort of thing would be trivial to cryptanalyze.

2. Row Transposition Ciphers

- A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns.
- The order of the columns then becomes the key to the algorithm.

For example,

Key: 3 4 2 1 5 6 7

Plaintext: a t t a c k p

o s t p o n e

d u n t i l t

w o a m x y z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

- A pure transposition cipher is easily recognized because it has the same letter frequencies as the original plaintext.
- For the type of columnar transposition just shown, cryptanalysis is fairly straightforward and involves laying out the ciphertext in a matrix and playing around with column positions.

- Diagram and trigram frequency tables can be useful.
- The transposition cipher can be made significantly more secure by performing more than one stage of transposition.
- The result is a more complex permutation that is not easily reconstructed.
- Thus, if the foregoing message is re encrypted using the same algorithm,

Key: 3 4 2 1 5 6 7

Input: t t n a a p t

m t s u o a o

d w c o i x k

n l y p e t z

Output: NSCYAUOPTTWLTMDNAOIEPAXTTOKZ

3. Rotor Machines

- The basic principle of the rotor machine.
- The machine consists of a set of independently rotating cylinders through which electrical pulses can flow.
- Each cylinder has 26 input pins and 26 output pins, with internal wiring that connects each input pin to a unique output pin.
- For simplicity, only three of the internal connections in each cylinder are shown.



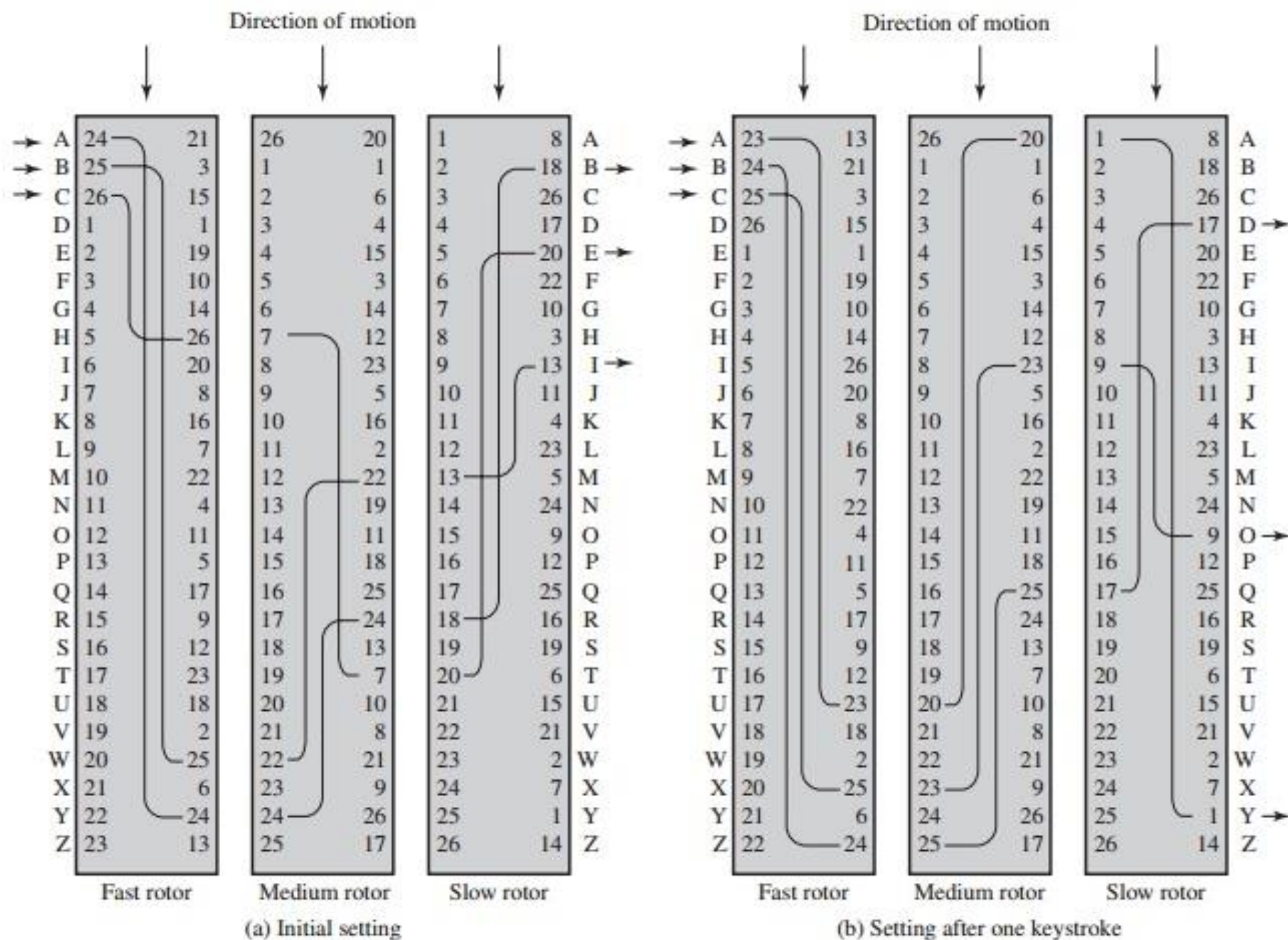


Figure 2.8 Three-Rotor Machine with Wiring Represented by Numbered Contacts

- If we associate **each input and output pin with a letter of the alphabet**, then a single cylinder defines a **monoalphabetic substitution**.
- For example, in Figure , if an operator depresses the key for the letter A, an electric signal is applied to the first pin of the first cylinder and flows through the internal connection to the twenty-fifth output pin.
- Consider a machine with a single cylinder. After each input key is depressed, the cylinder rotates one position, so that the internal connections are shifted accordingly. Thus, a different monoalphabetic substitution cipher is defined.
- **After 26 letters of plaintext**, the cylinder would be back to the initial position. Thus, we have a **polyalphabetic substitution** algorithm with a period of 26.
- The power of the rotor machine is in the use of multiple cylinders, in which the output pins of one cylinder are connected to the input pins of the next. Figure shows a three-cylinder system.

- The left half of the figure shows a position in which the input from the operator to the first pin (plaintext letter A) is routed through the three cylinders to appear at the output of the second pin (ciphertext letter B).
- With multiple cylinders, the one closest to the operator input rotates one pin position with each Key stroke.
- The right half of Figure shows the system's configuration after a single keystroke.
- For every complete rotation of the inner cylinder, the middle cylinder rotates one pin position. Finally, for every complete rotation of the middle cylinder, the outer cylinder rotates one pin position. This is the same type of operation seen with an odometer.
- The result is that there are $26 \times 26 \times 26 = 17,576$ different substitution alphabets used before the system repeats.
- The addition of fourth and fifth rotors results in periods of 456,976 and 11,881,376 letters, respectively

STEGANOGRAPHY

- A plaintext message may be hidden in one of two ways.
 - The methods of **steganography** conceal the existence of the message, whereas the methods of **cryptography** render the message unintelligible to outsiders by various transformations of the text.
 - A simple form of steganography, but one that is time-consuming to construct, is one in which an arrangement of words or letters within an apparently innocuous text spells out the real message.
- For example, the sequence of first letters of each word of the overall message spells out the hidden message.
- Figure shows an example in which a subset of the words of the overall message is used to convey the hidden message.

3rd March

Dear George,

Greetings to all at Oxford. Many thanks for your letter and for the Summer examination package. All Entry Forms and Fees Forms should be ready for final despatch to the Syndicate by Friday 20th or at the very latest, I'm told, by the 21st. Admin has improved here, though there's room for improvement still; just give us all two or three more years and we'll really show you! Please don't let these wretched 16t proposals destroy your basis O and A pattern. Certainly this sort of change, if implemented immediately, would bring chaos.

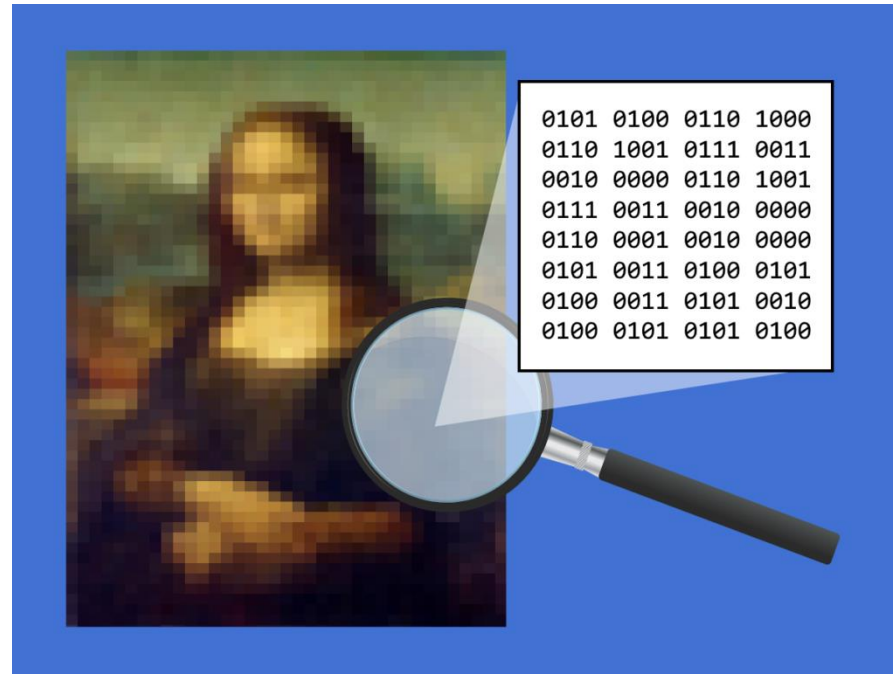
Sincerely yours.

Various other techniques have been used historically; some examples are the following:

- **Character marking:** Selected letters of printed or typewritten text are overwritten in pencil. The marks are ordinarily not visible unless the paper is held at an angle to bright light.
- **Invisible ink:** A number of substances can be used for writing but leave no visible trace until heat or some chemical is applied to the paper.
- **Pin punctures:** Small pin punctures on selected letters are ordinarily not visible unless the paper is held up in front of a light.
- **Typewriter correction ribbon:** Used between lines typed with a black ribbon, the results of typing with the correction tape are visible only under a strong light.

- Although these techniques may seem archaic, they have contemporary equivalents.
- [WAYN93] proposes hiding a message by using the least significant bits of frames on a CD.
- For example, the Kodak Photo CD format's maximum resolution is 2048 by 3072 pixels, with each pixel containing 24 bits of RGB color information. The least significant bit of each 24-bit pixel can be changed without greatly affecting the quality of the image.
- The result is that you can hide a 2.3-megabyte message in a single digital snapshot. There are now a number of software packages available that take this type of approach to steganography.
- Steganography has a number of **drawbacks** when compared to encryption.
- It requires a lot of overhead to hide a relatively few bits of information, although using some scheme like that proposed in the preceding paragraph may make it more effective.
- Also, once the system is discovered, it becomes virtually worthless. This problem, too, can be overcome if the insertion method depends on some sort of key .
- Alternatively, a message can be first encrypted and then hidden using steganography.

- The **advantage** of steganography is that it can be employed by parties who have something to lose should the fact of their secret communication (not necessarily the content) be discovered.
- Encryption flags traffic as important or secret or may identify the sender or receiver as someone with something to hide.



thank you